Rakesh James

AI Solution Architect · Agentic AI, Generative AI & Secure Cloud (AWS and Azure)

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Remote-first, UK and international · Contract or permanent · Immediate start

Profile

Hands-on AI Solution Architect with 17+ years in enterprise platforms, now focused on agentic and generative AI on AWS and Azure. I design the AI architecture and build it myself: multi-agent orchestration, agent runtimes, RAG pipelines, guardrails and evaluation, all delivered as infrastructure as code (Terraform, Bicep).

I prove routing and safety with live evaluations before cut-over, and I review agent identity and data isolation for security sign-off. Recent work covers a RAG AI Assistant and AI engineering agents for a UK energy-industry regulator, a multi-agent AI platform on Amazon Bedrock AgentCore, and a security review of a Microsoft 365 knowledge-agent swarm.

I work to regulated-delivery discipline from day one: one change per reviewed PR with plan evidence, production kept read-only, no secrets in code, least-privilege and time-boxed access.

Core expertise

Area	Skills and tools
Agentic AI	Multi-agent orchestration (hub-and-spoke), semantic routing, Model Context Protocol (MCP), Amazon Bedrock AgentCore Runtime, Claude Code agents, skills and hooks, approval-gated actions
Generative AI	RAG pipelines, embeddings and chunking, Amazon Bedrock (Converse, Guardrails, Knowledge Bases), Azure OpenAI, Azure AI Foundry
LLM evaluation	Live before/after routing measurement, replay harnesses with acceptance thresholds, safety-parity testing, grounding checks

Area	Skills and tools
Search and retrieval	Azure AI Search, OpenSearch Serverless and hybrid search, vector stores, Solr, Elasticsearch
AI security and identity	Agent identity, Microsoft Entra ID, managed identities, OBO token flows, Microsoft Graph permissions, Cognito, per-user and tenant data isolation, MCP security review
Cloud and infrastructure as code	AWS (Lambda, S3, EventBridge, CloudWatch, KMS, Cognito), Azure (Blob Storage, Logic Apps, Log Analytics, Power BI), Terraform, Bicep, Kubernetes, Docker
Delivery	Delivery and technical leadership, GitHub Actions, GitLab CI/CD, Azure DevOps, security scanning, E2E testing (Playwright, Nightwatch), WCAG 2.2 AA
Languages and platforms	Python, TypeScript / Node.js, PHP, JavaScript, React, Next.js, Drupal 6–11, Google Apigee, Bloomreach

Selected AI projects

RAG AI Assistant, Azure AI platform and AI engineering agents

Technical Lead & Solution Architect · UK energy-industry regulator

Led the architecture for moving the regulator's portal to Drupal 11 on Acquia Cloud, with a Microsoft Azure data and AI platform behind it and an AI-assisted engineering workflow built on Claude Code agents.

- **RAG search and AI Assistant:** chunks content and attachments (PDF, media, composite pages), embeds them with Azure OpenAI and indexes vectors in Azure AI Search, filtered by each user's group access; exclusion by URL or title with automatic cleanup in the index.
- **AI agents for the delivery team:** Claude Code agents, skills and hooks for merge-request review checked against code and linters, E2E test writing from Jira features, bulk Jira bug triage with fix MRs, Figma-to-code over MCP, and Confluence diagrams.
- **Agent guardrails and standards:** approval-gated writes, read-only production, hook-enforced safety, and AI engineering standards every agent follows.

- **MCP assessment:** assessed exposing the portal as an MCP server for AI clients and documented the security gaps to close first.
- **Azure platform:** infrastructure as code in Bicep (Azure DevOps); zero-secret Entra ID API security with managed identities and app-role mapping; Blob Storage, Logic Apps, Power BI with row-level security; Log Analytics audit retention for DPO sign-off.
- **Delivery:** 450+ migrations of content, organisations, users and regulatory data; GitLab CI/CD with tag-based deploys, pre-deploy backups, security scans and accessibility gates; New Relic observability.

Azure OpenAI, Azure AI Search, Entra ID, Log Analytics, Logic Apps, Power BI, Bicep, Azure DevOps, Claude Code, MCP, Drupal 11, Acquia Cloud, GitLab CI/CD

Multi-agent AI orchestration platform on Amazon Bedrock AgentCore

Hands-on Solution Architect / AI Platform Engineer · UK public sector (NDA) · 3 months

Moved a standalone citizen-facing GenAI assistant into a hub-and-spoke multi-agent orchestrator, where the hub routes each conversation to one of several specialist agents. Delivered in three phases and 21 pieces.

- **Runtime re-architecture:** moved the agent onto Amazon Bedrock AgentCore Runtime after Bedrock Agents closed to new accounts: a Python service on Converse, ApplyGuardrail and Knowledge Base Retrieve, with per-turn time budgets and explicit stop reasons.
- **External-runtime pattern:** designed how the orchestrator hosts agents that run in their own runtime; packaged the agent to the platform's agent standard as a versioned, pinned release.
- **RAG content pipeline as code:** a Terraform module for crawlers, refinery, LLM summariser, knowledge-base sync, scheduled crawls and alarms.
- **Semantic routing:** measured live routing before and after each change across 100 phrases, with no regression on neighbouring agents.
- **Evaluation and safety:** proved safety parity through the hub's safety stage; built a "not degraded" replay harness at a 90% acceptance threshold before cut-over.
- **Self-service content admin:** per-agent roles from token claims, S3 presigned uploads and caller-scoped limits.
- **Delivery:** ~85 commits and 20+ reviewed PRs, each infrastructure PR with its Terraform plan; production read-only throughout.

Amazon Bedrock AgentCore, Bedrock Guardrails and Knowledge Bases, OpenSearch Serverless, Terraform, Python, TypeScript, Lambda, EventBridge Scheduler, Cognito, S3, CloudWatch, KMS, GitHub Actions

Microsoft 365 knowledge-agent swarm: architecture and security review

Solution / Security Architect (review and assurance) · UK public sector (NDA) · 2 months

Independent architecture, access-control and security review of a workforce AI assistant: a gateway agent validates, routes and demotes tokens, then fans out to mail, calendar, OneDrive/SharePoint, Teams, knowledge (RAG) and graph sub-agents over MCP.

- **Architecture traced from the code:** components, request sequence, ingestion and sync, and trust boundaries, with four diagrams.
- **Identity and data isolation:** server-side Entra group checks, per-user and tenant ACL filters, and every delegated and application Microsoft Graph permission.
- **15 security findings** with severity, evidence and fixes, including one critical identity-takeover path and five high-severity issues, in a suggested remediation order.
- **Assurance gaps:** where the documented design disagrees with actual behaviour, and a stakeholder notes pack through 11 revisions.

MCP, Microsoft Graph (OBO), Entra ID, Cognito, AWS Lambda / SAM, API Gateway, OpenSearch, DynamoDB, Neptune, Bedrock

RAG patient-assistant copilot

Solution Architect · EU pharmaceutical group (NDA)

- RAG copilot on Azure OpenAI and AKS, with guardrails, PII handling and Entra ID access control.

Platform and integration projects

Sector	Project	Role
GB electricity system operator	API developer portal on Drupal, API models generated in Python and published to developers, Entra ID with SCIM provisioning, JavaScript map of the GB grid with live site status (6+ months)	Delivery Lead & Solution Architect
UK postal and parcels operator	Customer platform serving 31M+ UK addresses (React 11→19, Drupal 9→10, Entra ID CIAM); parcels platform Drupal 7→10 migration on Azure	Lead Developer / Drupal Architect

Sector	Project	Role
Global tea brand	Bloomreach and Drupal architecture, including a campaign game app built in Bloomreach (3 months)	Solution Architect
UK national museum	Drupal 8→10 upgrade for Cyber Essentials compliance, a React knowledge-search app, and a move to Solr 8 and Elasticsearch 8	Solution Architect
Global advertising agency network	Drupal 10 platform consolidating 30+ brand sites, with a component library in Tailwind, Storybook and Pattern Lab	Solution Architect
Global credit-reference agency	Public API developer portal for credit, identity and verification APIs on Google Apigee	API Solution Architect
US automotive retailer	Apigee developer portal with CI/CD-driven proxy creation, Ping ID and Jira automation	Lead Developer
International health organisation	Global nutrition-policy platform: Drupal 7→9, XMART DB and Power BI integration, team of 15	Lead Architect
International charity	17-site multilingual Acquia Site Factory estate, Stripe SCA payments, headless React micro-sites	Senior Developer
Global trade distributor	Enterprise Apigee developer portal with ADP SSO and custom Swagger UI	Lead Developer
Global media publisher	Unified Drupal 8 content platform for authors, editors and publishers	Lead Developer
New pharmaceutical brand	Greenfield Drupal 9 and Vue.js platform	Solution Architect

Career summary

17+ years from developer to architect: Software Programmer → Drupal Lead → Senior Drupal Consultant and Architect → Digital Architect for API developer portals → Drupal Lead Architect leading an EMEA CMS team → AI Solution Architect. Experience in the UK, EMEA and India across energy, public sector, pharma, financial services, retail, charity and global media.

Open source and community

- Top-30 global Drupal contributor (2015–2017), recognised in the official "Who Sponsors Drupal Development" reports.
- Author and maintainer of 15+ contributed modules, including Loqate, Custom Solr Search, Apigee App Dashboard, Page Access and Entity Reference Tab / Accordion Formatter.
- Speaker at DrupalCon and regional Drupal camps; trainer and mentor for teams in the UK and India.

Education

- B.Tech, Computer Science — Calicut University, India, 2007